Team ICB on FloorSet-Prime: RL tree builder + pair re-cut refinement (RPolyPlan / JigsawPlanner stages in progress)

10/21-10/23/2026 | all 100 test designs | updated 10/23

Summary

Flow	Weighted	Valid	HPWL	Area	V_rel	<=1.05	=1.000	HPWL<GT
Packer flow	1.2247	100/100	+0.102	+0.107	0.026	14	0	1
Packer + refine	1.1425	100/100	+0.035	+0.107	0.013	36	0	18
Partition (t80, cache fixed)	1.2915	99/100	+0.137	+0.000	0.041	18	0	2
Partition + refine	1.1462	99/100	+0.008	+0.000	0.014	69	39	45
Partition + refine in restarts	1.1229	99/100	+0.002	+0.000	0.007	81	48	50
Partition + refine + tree policy	1.1077	99/100	-0.022	+0.000	0.003	94	76	79
Final 10/22 night	1.0222	100/100	-0.010	+0.009	0.003	86	64	67
Final (best of all + polish)	1.0054	100/100	-0.028	+0.001	0.001	98	83	85
Best of all, picked per design	1.0054	100/100	-0.028	+0.001	0.001	98	83	85

Where the numbers come from: the final layout of each design is the best of several flows, then polished by the pair re-cut refinement (stage 3 of the RPolyPlan / JigsawPlanner plan, random kicks). The reinforcement-learning tree builder is in it: its flow ('RL' in the tables) gives 59 of the 100 final layouts. The RL kick policy is not used (it lost to random kicks), and stages 1-2 of the plan (global floorplan and grid assignment) are not used (their layouts were rarely valid).

RL tree builder against the same flow started from the seed tree (both with refinement in every restart, all 100 designs): better on 47 designs, worse on 4, equal on 48 (design 75 is invalid in both); mean cost 1.0069 vs 1.0234; wirelength 2.2% below the ground truth vs 0.2% above it; 76 designs at 1.000 vs 48.

Best layout per design over all flows: weighted cost 1.0054 on the 100 test designs, 100 valid; 83 designs at 1.000 (they match or beat the ground truth), 85 with shorter wirelength than the ground truth, 98 within 5%. The 10/20 report's best of both was 1.160.

The partition flow alone (cache fixed, before refinement) scores 1.2915; the pair re-cut refinement accounts for most of the gain over it.

The plan

Prime's ground truth tiles its frame exactly (zero white space) with rectilinear blocks, and 85 of the 100 test layouts cannot be built from straight cuts alone. The plan agreed on 10/21 combines two published rectilinear floorplanners with reinforcement learning (RL), inside a mix of methods whose best layout is picked per design:

1. Global floorplan in the style of RPolyPlan (arXiv 2610.28276): an augmented-Lagrangian optimization of soft-block boxes with pairwise overlap sharing, inside the exact frame.

2. Legalization in the style of JigsawPlanner (ICCAD 2024): Prime's coordinates and areas are integers, so unit grid cells can give every block its exact area with no white space and no overlap.

3. Clean-up moves that keep areas exact, chosen by an RL policy instead of hand-written rules.

Alongside: (A) an RL policy that builds the slicing tree for the existing zero-white-space partition flow.

What was built overnight, and what worked

Regression fix. A cache cap added on 10/21 to share the machine with Vivado made the partition search clear its cache constantly; big designs failed or lost 0.3-0.6. The old size is the default again (about 6 GB for 6 workers), and the full partition run was redone with it ('Partition' column).

Stages 1-2 (grid flow, prime_grid.py). The augmented-Lagrangian global floorplan and the exact grid assignment (a transportation linear program solved with HiGHS) both work end to end, and their wirelength is good (about 6% over the ground truth). But stage 1 does not converge at 0% white space, the cell assignment leaves staircase edges that break the 16-corner cap, and clusters split, so few layouts came out valid. This flow is parked; its parts are reusable.

Stage 3 on the partition flow (rl/refine.py) -- the night's main gain. Instead of legalizing a grid, the stage-3 moves were applied to the partition flow's already-valid layouts: two touching soft blocks are merged and re-split by one straight cut at the exact area of each; a rigid block and a soft neighbour swap places within their joint region; three soft blocks are re-split by two cuts. Areas stay exact and white space stays zero. A move is kept if it lowers a stand-in for the contest cost (wirelength and soft violations). Pairs are re-cut independently of the tree, so layouts leave the slicing family. Greedy descent is followed by iterated local search: a few random re-cuts ('kicks'), descend again, keep only improvements.

RL in stage 3 (rl/kick_policy.py). A small network scores candidate re-cuts from local features (immediate gain, distance of each block from where its nets pull it, net weight between the pair, soft violations) and samples the kicks of the iterated local search; REINFORCE, comparing several kicks from the same state. v1 (111 training designs, re-scoring every candidate after each move) tied with random kicks on 5 held-out designs; v2 (one scoring pass over 60 sampled pairs per kick, trained from a pool of 150 training layouts) lost on 14 of 19 held-out designs (mean 1.018 vs 1.011 for random kicks at equal time). Random kicks stay in use.

RL tree builder (A, rl/tree_policy.py). The partition annealer used to start from a tree that is invalid on most large designs (41-51 bad leaves on test designs 60 and 90), mostly because rigid blocks ended up as siblings. The policy builds the tree top-down over the real region, masks splits that leave a rigid-only subtree or a bad leaf, and learns sort keys, cut directions and split points by REINFORCE against the mean of 8 sampled trees per design (best validation tree cost 8.56, against 21.7 for the old first tree). Each restart starts its search from the best of 32 sampled trees. Its first A/B looked like a loss (2 of 12 layouts invalid), but both failures came from the refiner's polygon output (1e-6 coordinate jogs, a collinear vertex), fixed on 10/22 together with a guard that checks every refined layout. A first full run then crashed on 17 designs from a refiner sliver bug, and the crash also switched the restarts to serial; both were fixed and those designs rerun.

Bugs fixed on 10/22

Refiner output: coordinates closer than 1e-5 are merged (rigid blocks keep their exact values), collinear vertices removed, cuts leaving pieces thinner than 0.001 rejected; every refined layout is checked against the evaluator's hard rules, with the unrefined one as fallback.

Evaluator shape check (rl/polygon.py canonical_key): float noise from a fractional position left a duplicate vertex, so an exact fixed shape read as different; the ground truth still scores 1.000 on all 100 designs.

A crashing restart made the restart pool fall back to serial for the rest of the run (4x slower); restarts now catch their own errors.

Targeted pushes for edge-pin misses and split clusters (rl/refine.py fix_violations): fixed designs 66 and 70 (to 1.000); blocks walled in by preplaced or rigid blocks (designs 50, 80, 81, 93, 98) still miss their edge.

Fixes on 10/23

Design 75 (1.289 -> 1.003): a 5 x 7 pocket walled in by preplaced blocks could only be filled by a member of a soft MIB group of area 35, but MIB groups got a fixed 1.5-aspect rectangle that never fit. A rectangular pocket now gives a matching MIB group its shape (rl/partition.py _fill_pockets).

Design 77 (1.157 -> 1.041): the solve-time pick (prime_evaluate.relative_cost, and rl/repack.py contest_relative_cost) added ratios where the contest adds gaps; the base near 2 doubled the weight of soft violations and chose the packer layout over a better partition one. With gaps the pick equals picking with the ground truth (1.010 both). The original Lite contest repo has the same function and was not changed.

Edge pins and clusters behind walls: a beam-search 'route push' lets a block go around preplaced or rigid blocks (it fixed the split cluster of design 81, 1.044 -> 1.005); designs 50, 77, 80 and 98 have no valid sequence of pair moves to their edge and need a new move (e.g. swapping with a block already on the edge). A shape check now rejects moves that leave a self-touching ring (design 80).

Grid flow: starting stage 1 with 10% slack and shrinking it to zero does not help; it stalls at the same violation, so its formulation needs rework.

RL tree builder with a reward closer to its use (cost after a short anneal from each sampled tree, 6 designs in parallel): no gain on unseen designs. Round 1 (1.5 s timed anneals) lost a fair paired test, worse on 18 of 30 held-out designs; the timed anneal made the reward follow machine load. Round 2 (3,000-step deterministic anneals) improved its 30 validation designs (cost ratio to the seed tree 1.003 -> 0.889) but tied the current policy on 30 fresh held-out designs (15 better, 15 worse; mean 3.514 vs 3.458): the checkpoint choice fitted the validation set. The 10/21 policy stays in use. A real gain likely needs far more updates than the 101-355 a 1.5-hour run allows.

Mean cost by design size

Blocks	Designs	Packer	Pack+R	Partition	Part+R	Part+R2	RL	Night	Final	Best
21-40	20	1.106	1.061	1.139	1.047	1.025	1.006	1.008	1.002	1.002
41-60	20	1.130	1.089	1.106	1.024	1.015	1.001	1.005	1.000	1.000
61-80	20	1.161	1.089	1.151	1.031	1.021	1.005	1.011	1.004	1.004
81-100	20	1.212	1.128	1.694	1.507	1.480	1.461	1.038	1.004	1.004
101-120	20	1.226	1.153	1.189	1.048	1.025	1.011	1.021	1.006	1.006

All 100 designs

Quality cost per design (the contest cost without its runtime term). 1.000 is the ground truth's own score and also the floor: gaps below zero count as zero, so a design at 1.000 matches or beats the ground truth. 'HPWL gap' is the best layout's wirelength relative to the ground truth (negative = shorter). 'Picked' is the flow the best-of chose without looking at the ground truth.

ID	Blocks	Packer	Pack+R	Partition	Part+R	Part+R2	RL	Night	Final	Best	HPWL gap	Picked
0	21	1.168	1.021	1.177	1.000	1.000	1.000	1.000	1.000	1.000	-0.052	RL
1	22	1.045	1.045	1.041	1.000	1.000	1.000	1.000	1.000	1.000	-0.251	Part+R2
2	23	1.064	1.041	1.054	1.000	1.000	1.000	1.000	1.000	1.000	-0.032	Part+R
3	24	1.084	1.062	1.190	1.125	1.000	1.000	1.000	1.000	1.000	-0.073	RL
4	25	1.098	1.061	1.098	1.011	1.016	1.000	1.000	1.000	1.000	-0.065	RL
5	26	1.112	1.109	1.632	1.632	1.172	1.000	1.108	1.000	1.000	-0.038	RL
6	27	1.035	1.017	1.024	1.000	1.000	1.000	1.000	1.000	1.000	-0.017	Part+R2
7	28	1.138	1.021	1.183	1.000	1.000	1.000	1.000	1.000	1.000	-0.102	RL
8	29	1.228	1.027	1.590	1.125	1.125	1.125	1.027	1.027	1.027	-0.105	Pack+R
9	30	1.086	1.053	1.021	1.000	1.000	1.000	1.000	1.000	1.000	-0.050	Part+R
10	31	1.065	1.039	1.074	1.005	1.000	1.000	1.000	1.000	1.000	-0.034	RL
11	32	1.036	1.020	1.064	1.000	1.000	1.000	1.000	1.000	1.000	-0.071	Part+R
12	33	1.179	1.155	1.071	1.000	1.122	1.000	1.000	1.000	1.000	-0.022	RL
13	34	1.145	1.220	1.097	1.000	1.000	1.000	1.000	1.000	1.000	-0.045	Part+R
14	35	1.045	1.020	1.087	1.000	1.000	1.000	1.000	1.000	1.000	-0.152	Part+R2
15	36	1.065	1.056	1.065	1.021	1.026	1.004	1.021	1.004	1.004	+0.008	RL
16	37	1.088	1.055	1.114	1.013	1.000	1.000	1.000	1.000	1.000	-0.065	RL
17	38	1.046	1.025	1.024	1.003	1.000	1.000	1.000	1.000	1.000	-0.009	Part+R2
18	39	1.172	1.150	1.045	1.000	1.008	1.000	1.000	1.000	1.000	-0.027	Part+R
19	40	1.227	1.026	1.134	1.009	1.023	1.000	1.000	1.000	1.000	-0.035	RL
20	41	1.031	1.013	1.009	1.000	1.000	1.000	1.000	1.000	1.000	-0.031	Part+R2
21	42	1.024	1.010	1.036	1.000	1.000	1.000	1.000	1.000	1.000	-0.063	Part+R
22	43	1.050	1.037	1.047	1.000	1.000	1.000	1.000	1.000	1.000	-0.025	RL
23	44	1.078	1.034	1.021	1.000	1.000	1.000	1.000	1.000	1.000	-0.074	Part+R2
24	45	1.306	1.263	1.296	1.108	1.020	1.000	1.012	1.000	1.000	-0.019	RL
25	46	1.129	1.101	1.123	1.002	1.038	1.004	1.000	1.000	1.000	-0.008	Part+R
26	47	1.265	1.208	1.101	1.008	1.002	1.000	1.000	1.000	1.000	-0.017	Part+R2
27	48	1.097	1.047	1.077	1.006	1.073	1.000	1.000	1.000	1.000	-0.008	Part+R
28	49	1.310	1.170	1.076	1.021	1.000	1.000	1.000	1.000	1.000	-0.054	RL
29	50	1.114	1.105	1.222	1.008	1.006	1.002	1.003	1.001	1.001	+0.002	RL
30	51	1.053	1.015	1.051	1.000	1.000	1.000	1.000	1.000	1.000	-0.029	Part+R2
31	52	1.084	1.038	1.045	1.000	1.000	1.000	1.000	1.000	1.000	-0.082	Part+R

32	53	1.108	1.063	1.069	1.069	1.000	1.000	1.000	1.000	1.000	-0.104	Part+R2
33	54	1.081	1.025	1.080	1.000	1.000	1.000	1.000	1.000	1.000	-0.101	Part+R
34	55	1.155	1.117	1.216	1.079	1.024	1.000	1.013	1.000	1.000	-0.022	RL
35	56	1.134	1.114	1.062	1.004	1.000	1.000	1.000	1.000	1.000	-0.033	RL
36	57	1.170	1.163	1.349	1.090	1.064	1.012	1.060	1.007	1.007	+0.013	RL
37	58	1.127	1.027	1.019	1.000	1.000	1.000	1.000	1.000	1.000	-0.033	RL
38	59	1.155	1.118	1.114	1.073	1.001	1.000	1.000	1.000	1.000	-0.014	RL
39	60	1.138	1.118	1.103	1.003	1.072	1.000	1.003	1.000	1.000	-0.002	RL
40	61	1.061	1.047	1.036	1.003	1.000	1.000	1.000	1.000	1.000	-0.007	RL
41	62	1.313	1.021	1.054	1.000	1.000	1.000	1.000	1.000	1.000	-0.067	Part+R
42	63	1.070	1.028	1.059	1.000	1.000	1.000	1.000	1.000	1.000	-0.049	RL
43	64	1.193	1.095	1.282	1.004	1.000	1.000	1.000	1.000	1.000	-0.026	RL
44	65	1.049	1.039	1.004	1.000	1.000	1.000	1.000	1.000	1.000	-0.011	RL
45	66	1.040	1.019	1.093	1.000	1.000	1.000	1.000	1.000	1.000	-0.025	Part+R
46	67	1.033	1.015	1.021	1.000	1.000	1.000	1.000	1.000	1.000	-0.017	RL
47	68	1.135	1.127	1.189	1.116	1.003	1.000	1.001	1.000	1.000	-0.006	RL
48	69	1.089	1.063	1.322	1.088	1.229	1.034	1.063	1.028	1.028	+0.056	RL
49	70	1.355	1.290	1.246	1.021	1.011	1.002	1.005	1.000	1.000	-0.014	RL
50	71	1.494	1.236	1.263	1.192	1.121	1.057	1.118	1.057	1.057	-0.018	RL
51	72	1.221	1.156	1.171	1.000	1.002	1.000	1.000	1.000	1.000	-0.011	RL
52	73	1.447	1.260	1.104	1.045	1.000	1.000	1.000	1.000	1.000	-0.034	Part+R2
53	74	1.072	1.017	1.134	1.000	1.000	1.000	1.000	1.000	1.000	-0.041	RL
54	75	1.096	1.055	1.110	1.067	1.000	1.000	1.000	1.000	1.000	-0.025	Part+R2
55	76	1.034	1.021	1.019	1.000	1.002	1.000	1.000	1.000	1.000	-0.014	RL
56	77	1.222	1.156	1.148	1.001	1.011	1.000	1.000	1.000	1.000	-0.041	RL
57	78	1.072	1.036	1.247	1.037	1.018	1.000	1.012	1.000	1.000	-0.021	RL
58	79	1.076	1.041	1.214	1.000	1.000	1.000	1.000	1.000	1.000	-0.033	RL
59	80	1.141	1.060	1.309	1.048	1.019	1.000	1.014	1.000	1.000	-0.016	RL
60	81	1.174	1.083	1.025	1.008	1.005	1.000	1.003	1.000	1.000	-0.008	RL
61	82	1.031	1.020	1.181	1.047	1.001	1.000	1.000	1.000	1.000	-0.009	Part+R2
62	83	1.242	1.207	1.172	1.139	1.000	1.000	1.000	1.000	1.000	-0.011	Part+R2
63	84	1.149	1.106	1.687	1.197	1.189	1.004	1.106	1.001	1.001	+0.002	RL
64	85	1.322	1.059	1.222	1.080	1.019	1.000	1.013	1.000	1.000	-0.013	RL
65	86	1.186	1.086	1.057	1.000	1.000	1.000	1.000	1.000	1.000	-0.045	Part+R2
66	87	1.212	1.179	1.345	1.043	1.043	1.000	1.043	1.000	1.000	-0.031	RL
67	88	1.191	1.094	1.187	1.047	1.030	1.000	1.016	1.000	1.000	-0.026	RL
68	89	1.299	1.141	1.250	1.089	1.003	1.000	1.003	1.000	1.000	-0.007	RL
69	90	1.136	1.105	1.091	1.002	1.000	1.000	1.000	1.000	1.000	-0.006	Part+R2
70	91	1.329	1.283	1.275	1.091	1.070	1.000	1.070	1.000	1.000	-0.019	RL

71	92	1.262	1.146	1.464	1.062	1.013	1.000	1.002	1.000	1.000	-0.022	RL
72	93	1.136	1.068	1.025	1.000	1.000	1.000	1.000	1.000	1.000	-0.010	RL
73	94	1.143	1.120	1.416	1.000	1.000	1.080	1.000	1.000	1.000	-0.022	Part+R
74	95	1.204	1.110	1.151	1.000	1.000	1.000	1.000	1.000	1.000	-0.032	RL
75	96	1.332	1.291	10.000	10.000	10.000	10.000	1.290	1.003	1.003	+0.007	Pocket fix
76	97	1.058	1.042	1.372	1.211	1.052	1.047	1.041	1.041	1.041	+0.026	Pack+R
77	98	1.181	1.158	1.400	1.098	1.099	1.083	1.157	1.041	1.041	+0.005	RL
78	99	1.282	1.222	1.491	1.028	1.044	1.004	1.015	1.000	1.000	-0.007	RL
79	100	1.361	1.043	1.067	1.000	1.043	1.000	1.000	1.000	1.000	-0.008	Part+R
80	101	1.364	1.299	1.447	1.080	1.060	1.085	1.056	1.054	1.054	+0.033	Part+R2
81	102	1.251	1.219	1.268	1.104	1.060	1.046	1.051	1.005	1.005	+0.010	RL
82	103	1.074	1.042	1.036	1.000	1.000	1.000	1.000	1.000	1.000	-0.032	Part+R
83	104	1.252	1.231	1.229	1.089	1.018	1.018	1.013	1.013	1.013	+0.025	Part+R2
84	105	1.139	1.095	1.089	1.000	1.000	1.000	1.000	1.000	1.000	-0.034	Part+R
85	106	1.355	1.326	1.108	1.002	1.001	1.000	1.000	1.000	1.000	-0.008	Part+R2
86	107	1.184	1.149	1.319	1.096	1.070	1.008	1.070	1.004	1.004	+0.007	RL
87	108	1.142	1.102	1.406	1.140	1.072	1.003	1.100	1.000	1.000	+0.001	RL
88	109	1.242	1.163	1.156	1.007	1.011	1.000	1.004	1.000	1.000	-0.004	RL
89	110	1.211	1.064	1.077	1.059	1.000	1.000	1.000	1.000	1.000	-0.008	Part+R2
90	111	1.232	1.180	1.078	1.010	1.000	1.000	1.000	1.000	1.000	-0.026	RL
91	112	1.237	1.066	1.113	1.000	1.003	1.000	1.000	1.000	1.000	-0.007	Part+R
92	113	1.050	1.030	1.131	1.000	1.000	1.000	1.000	1.000	1.000	-0.016	Part+R
93	114	1.243	1.140	1.187	1.059	1.062	1.000	1.053	1.000	1.000	-0.018	RL
94	115	1.040	1.023	1.084	1.000	1.000	1.000	1.000	1.000	1.000	-0.032	RL
95	116	1.693	1.369	1.270	1.045	1.045	1.000	1.005	1.000	1.000	-0.014	RL
96	117	1.204	1.094	1.055	1.000	1.000	1.000	1.000	1.000	1.000	-0.021	RL
97	118	1.203	1.126	1.339	1.051	1.016	1.037	1.013	1.012	1.012	+0.024	Part+R2
98	119	1.289	1.264	1.124	1.083	1.083	1.033	1.053	1.029	1.029	+0.006	RL
99	120	1.122	1.072	1.268	1.125	1.000	1.000	1.000	1.000	1.000	-0.042	RL

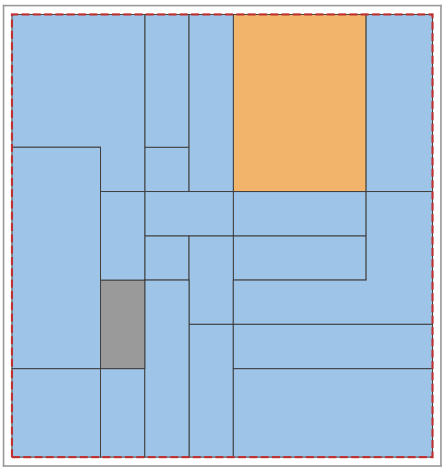
Anchor designs

The five anchor designs (0, 24, 49, 74 and 98, spread over the size range) drawn at one scale: the ground truth, the partition flow before refinement, and the final layout (green frame).

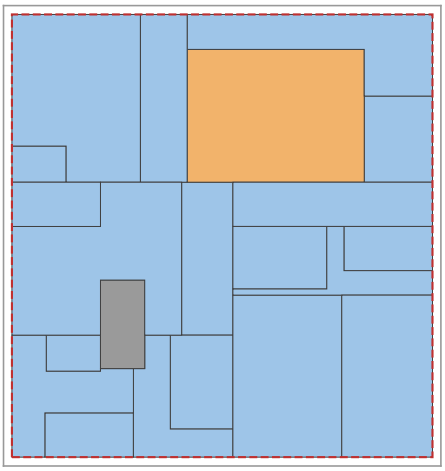
soft block fixed shape preplaced red dashes: W x H frame

Design 0 (21 blocks): Partition 1.177 | best 1.000

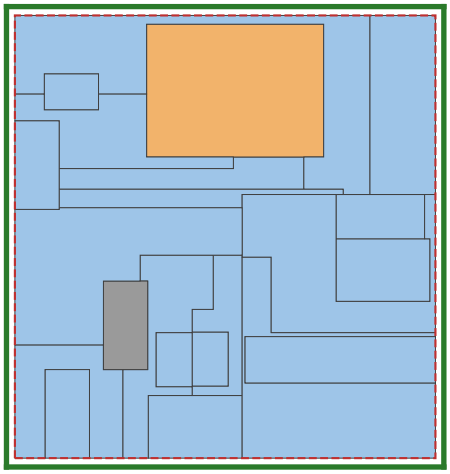
Ground truth (1.000)



Partition (1.177)

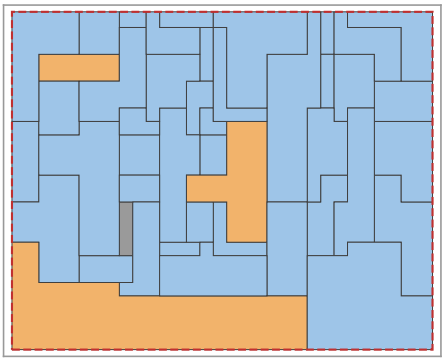


Final, from RL (1.000) [picked]

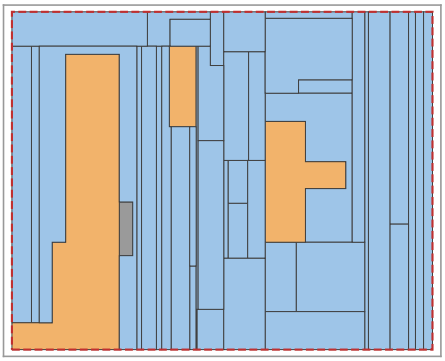


Design 24 (45 blocks): Partition 1.296 | best 1.000

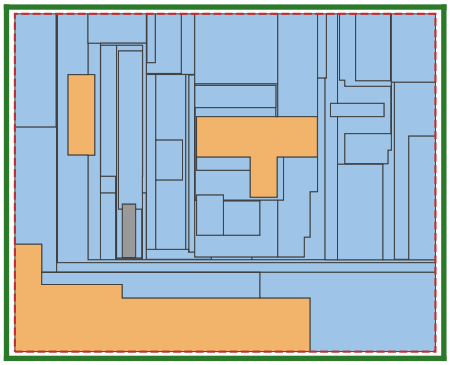
Ground truth (1.000)



Partition (1.296)

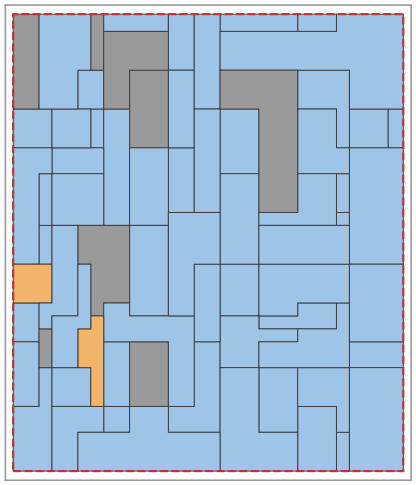


Final, from RL (1.000) [picked]

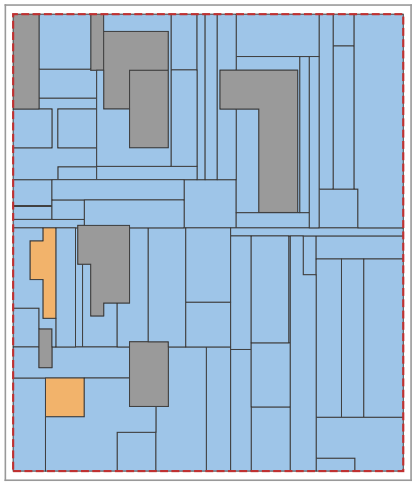


Design 49 (70 blocks): Partition 1.246 | best 1.000

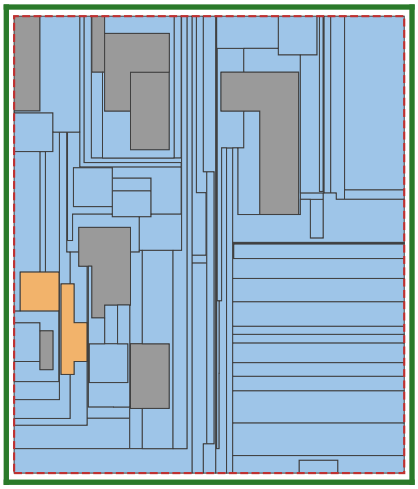
Ground truth (1.000)



Partition (1.246)

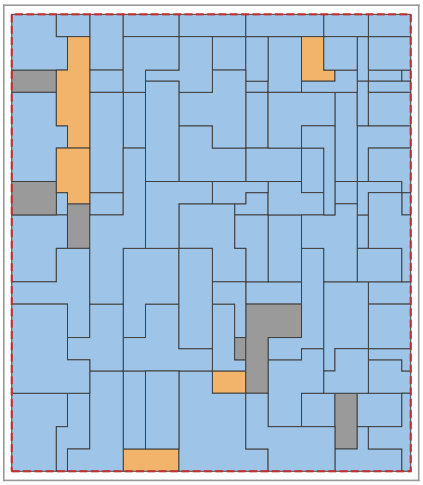


Final, from RL (1.000) [picked]

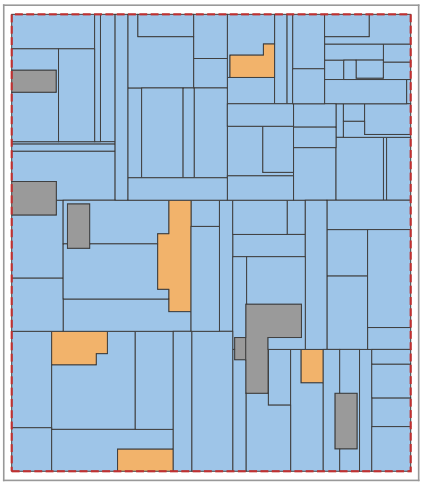


Design 74 (95 blocks): Partition 1.151 | best 1.000

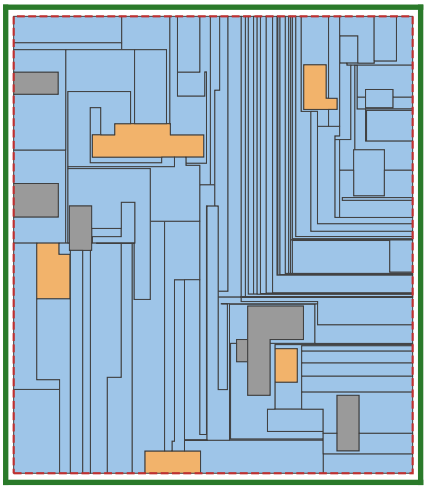
Ground truth (1.000)



Partition (1.151)

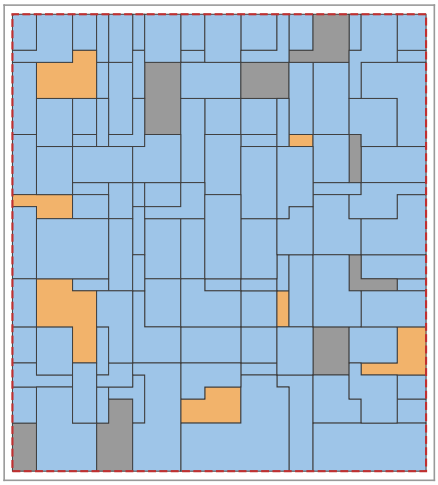


Final, from RL (1.000) [picked]

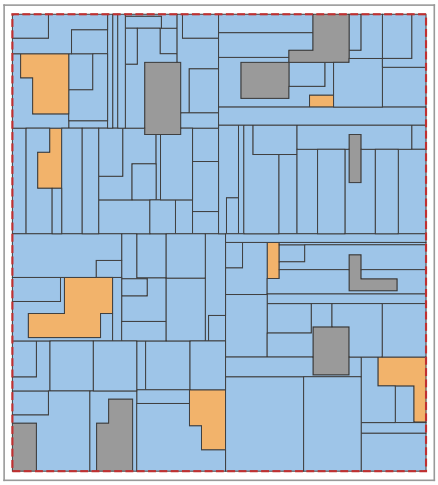


Design 98 (119 blocks): Partition 1.124 | best 1.029

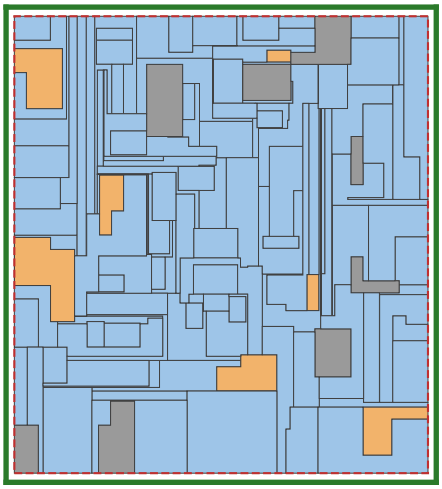
Ground truth (1.000)



Partition (1.124)



Final, from RL (1.029) [picked]



Next steps

1. Edge pins on designs 50, 77, 80 and 98: a move that swaps a block with one already on the required edge, neighbours absorbing the area difference.

2. RL tree builder: longer training with the fixed-step anneal reward and a larger held-out set for choosing checkpoints.

3. The grid flow (stages 1-2): reformulate the global floorplan's area sharing so it converges at zero white space.

4. Port the pick-rule fix (gaps, not ratios) to the Lite contest repo if you want it there.

5. Runtime: refinement is anytime; the contest budget decides how long it runs.